Classical and Self-Supervised Hankel Filtering for Magnitude-Only Rydberg Channel Estimation

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***Abstract*—Rydberg atomic receivers can estimate wireless channels from signal magnitudes measured with a known reference. Channel structure can help, but too much filtering can remove useful signal components. We study classical and learned ways to control Hankel filtering under the same nonlinear measurement model. A small estimator learns smooth filtering from withheld measurements, without true-channel training targets. We compare it with hard-rank methods and with OptShrink using both offline tuning and measurement-based rank selection. The compact estimator improves average cellwise accuracy over offline-tuned OptShrink on a broad tested population. Allowing OptShrink to select rank from each received block reverses that ranking, at higher processing cost. The learned method has fewer errors above the stated severe-error threshold, but does not dominate the error distribution. Weak references and nearly noiseless signals expose substantial limits. The result is a conditional accuracy–cost trade-off, not a universally better shrinker or a guarantee of reliable recovery.**

***Index Terms*—Channel estimation, Hankel matrices, magnitude measurements, Rydberg receivers, self-supervised learning.**

I. Introduction

RYDBERG atomic receivers sense radio-frequency electric fields through highly excited atoms. Array models make it possible to study multiuser communication with these receivers [1], [2]. In the magnitude-only setting considered here, the received signal combines with a known reference before its amplitude is measured. Estimating the complex channel is then a nonlinear inverse problem. The reference provides phase information indirectly, but noise, pilot design and initialization still affect recovery.

A geometric channel also has structure across the array. A small number of propagation paths produces a sum of spatial exponentials, whose Hankel matrix has low rank. Filtering this matrix can remove noise, but can also erase weak paths. Choosing a rank separately for every received block adds work. This motivates a practical question: can a small learned rule choose useful filtering at low cost, and when does a well-tuned classical method remain better?

We make three contributions. First, we measure how array size, channel structure and repeated hard-rank filtering affect estimation. Second, we develop a 188-parameter estimator that learns smooth filtering from withheld measurements while keeping the exact Rician update. Third, we compare it with tuned OptShrink and measurement-based rank selection on paired observations, with separate reporting of accuracy, large errors and processing time. The broad comparisons retain the original trained models and use fresh test data. Earlier classical wins and failed extensions are retained. The contribution is this compact estimator and its measured operating limits, not a new claim about low-rank filtering itself.

A. Relation to Existing Methods

EM-GS already exploits the referenced magnitude model in atomic MIMO reception [1]. Xu et al. study channel estimation using a strong-reference linearization [3]; our experiments keep exact magnitude observations. URformer already unfolds phase retrieval for Rydberg receivers [4]. Thus neither the observation model nor unfolding alone is a new contribution. Quantum-MUSIC uses recovered channel information for direction finding [5]. A recent robust variant studies outlier-contaminated observations [6]; our Gaussian-field-noise tests do not establish that kind of robustness.

Alternating rank and structure projections are established [7], as are low-rank Hankel recovery [8], [9], structured alternating least squares [10] and data-driven singular-value shrinkage [11]. Learned Hankel reconstruction [12], self-supervision through withheld measurements [13], [14], and Rician-likelihood training [15] also have precedents. Self-supervised denoising has been studied for Rydberg sensor signals [16]. Our narrower contribution is to learn a small spatial correction within exact referenced-magnitude channel estimation and assess its value against these structural alternatives. We do not claim the first use of any of these ingredients

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or superiority over published URformer, for which a matched comparison is not available here.

Other relevant methods make different structural assumptions. LRPR-VBL uses a hierarchical low-rank matrix prior and variational EM for magnitude-only recovery [17]. HVaF exploits a Vandermonde factorization to recover exponential signals from incomplete samples [18]. LRHD studies low-rank Hankel denoising [9]. Our in-loop OptShrink control is not a reproduction of those full methods. Adapting them to the known-reference Rician model, with independent parameter selection, remains an empirical comparison gap. Their published denoising or recovery guarantees are not transferred to our nonlinear reconstruction.

II. Observation Model and Available Information

An array with N elements receives P pilots from K users. Its observations are

$$Z = |GS + B + W|, \qquad W_{np} \sim \mathcal{CN}(0, v), \quad (1)$$

where $G \in \mathbb{C}^{N\times K}$ is the unknown channel, $S \in \mathbb{C}^{K\times P}$ contains known pilots, and B is a known reference field. Magnitudes act entrywise. Noise samples are independent, with total complex variance v and variance $v/2$ in each real component. Receiver conversion gain is absorbed into the scaling. This effective measurement model does not simulate atomic dynamics or optical readout.

For a half-wavelength uniform linear array, each user channel is

$$g_{nk} = \sum_{\ell=1}^{L_k} \alpha_{\ell k} e^{-j\pi n \sin\theta_{\ell k}}, \qquad n = 0, \dots, N-1. \quad (2)$$

Unless specified otherwise, $\alpha_{\ell k} \sim \mathcal{CN}(0, 1/L_k)$ and angles are independent and uniform on $(-\pi/2, \pi/2)$. Pilots follow $\mathcal{CN}(0,1)$. The simulated reference is a broadside field with constant waveform and zero phase,

$$B_{np} = 10^{\mathrm{RSR}/20}, \qquad v = K10^{-\mathrm{SNR}/10}. \quad (3)$$

SNR is relative to aggregate user power, whereas RSR is relative to one user's power. At $K = 3$, RSR 12 dB corresponds to 7.23 dB relative to total signal power. Channels have unit expected power per user; individual realizations are not rescaled to identical energy. Reference calibration and noise variance are assumed known in the main comparisons.

A. Rician Likelihood and Local Identifiability

For $a = |\mu|$, a single observation has density

$$p(z \mid a, v) = \frac{2z}{v} e^{-(z^2+a^2)/v} I_0(2za/v), \quad z \ge 0. \quad (4)$$

With fixed v, a stable negative log-likelihood, up to observation-only terms, is

$$\ell(z, \mu, v) = \frac{(z - |\mu|)^2}{v} - \log I_{0e}(2z|\mu|/v), \quad (5)$$

where $I_{0e}(x) = e^{-x} I_0(x)$ for $x \ge 0$. Scaled Bessel functions avoid overflow at large arguments. If variance is estimated, the omitted $\log v$ term must be restored.

For one array row, let $\mu_p = g^T s_p + b_p$ and $x = [\mathrm{Re}\, g^T, \mathrm{Im}\, g^T]^T$. Where $\mu_p \neq 0$, the real Jacobian of its noiseless magnitudes has row

$$A_p = [\mathrm{Re}\, c_p^T, -\mathrm{Im}\, c_p^T], \qquad c_p = \mu_p^* s_p / |\mu_p|. \quad (6)$$

If A has rank $2K$, a nonsingular square sub-Jacobian and the inverse function theorem give local recovery. Therefore $P \ge 2K$ is necessary for this regular full-rank condition, but is not sufficient. A small minimum singular value means that recovery can still be unstable. This is a local statement for an unrestricted channel row, not a global or sparse-channel recovery guarantee [19].

With $B = 0$, a common phase rotation of g leaves its magnitudes unchanged. A known nonzero reference can break this ambiguity, but need not do so for every pilot/reference design. For example, if $b_p = c^T s_p$, then rotating $g + c$ leaves the observations unchanged. Unknown common reference phase also permits simultaneous rotation of channel and reference: $|e^{j\phi} GS + e^{j\phi} B| = |GS + B|$. Estimating an unrestricted common phase jointly with G cannot fix the absolute phase from these observations alone. An additional phase anchor or other information is required. The known-reference assumption is therefore substantive, not just a normalization.

B. Information References

For independent measurements, the real Fisher information is

$$J = A^T \,\mathrm{diag}(w_p) A,$$
$$w_p = \mathbb{E}\left[\left(-\frac{2a_p}{v} + \frac{2Z_p}{v}\frac{I_1(2Z_p a_p/v)}{I_0(2Z_p a_p/v)}\right)^2\right]. \quad (7)$$

The full channel information combines array rows in the corresponding real-coordinate ordering. One amplitude observation contributes an outer product and hence at most one information direction. Several different pilots and the known reference are needed to distinguish the complex channel coordinates.

Where the information matrices are nonsingular, the unconstrained reference is J^{-1}. The geometric constrained Cramér–Rao bound (CCRB) is [20]

$$C_{\mathrm{geom}} = U(U^T J U)^{-1} U^T. \quad (8)$$

Here U is an orthonormal basis for the channel manifold's real tangent space at the true path configuration. It is constructed from derivatives with respect to

real and imaginary path gains and spatial frequencies. With $\omega = \pi \sin\theta$, the complex derivative columns are $e^{-jn\omega}$, $je^{-jn\omega}$ and $-jn\alpha e^{-jn\omega}$. Stacking their real and imaginary parts defines the tangent space; gains and frequencies remain unknown parameters. The true model order is used only for this ideal reference.

We normalize the summed traces by the same summed channel energy used for NMSE. The gap between the two bounds describes the value of geometric information under local unbiasedness assumptions. It is not guaranteed attainable improvement. Our estimators can be biased, so these curves are not absolute lower limits on their measured NMSE.

III. Classical Hankel Estimation

A. A Common Rician Measurement Update

Let $M = GS+B$ and $\kappa = 2Z|M|/v$. The conditional complex field and EM update are

$$Q = Z \odot \frac{I_1(\kappa)}{I_0(\kappa)} \odot \frac{M}{|M|}, \tag{9}$$

$$G_{\mathrm{EM}} = (Q-B)S^H(SS^H)^{-1}. \tag{10}$$

All ratios in the first line act entrywise. At $M = 0$, the continuous limit is zero; no substantial amplitude floor changes the likelihood. Linear solves evaluate the second line, requiring full row rank of the fitting pilot matrix.

For $\mathcal{L}(G) = \sum_{np} \ell(Z_{np}, [GS+B]_{np}, v)$, its Wirtinger derivative is

$$D = \frac{\partial \mathcal{L}}{\partial G^*} = \frac{(M-Q)S^H}{v},$$

$$G_{\mathrm{EM}} = G - vD(SS^H)^{-1}. \tag{11}$$

Thus EM is an exact preconditioned gradient update for this likelihood. The corresponding packed real gradient is $2D$. This distinction fixes the factor of two when comparing gradient descent and EM-based unfolding.

All iterative comparisons start from spectral initialization. For row n, form C_n with rows $[s_p^T, b_{np}]$, let u_n be the leading eigenvector of $C_n^H \operatorname{diag}(z_n) C_n$, and set

$$c_n = \frac{\sum_p z_{np} |(C_n u_n)_p|}{\sum_p |(C_n u_n)_p|^2}. \tag{12}$$

The first K entries of $c_n u_n e^{-j \arg(u_{n,K+1})}$ initialize the channel row. Whenever pilots are withheld, initialization uses only the fitting subset. This rule is necessary to avoid leaking validation observations into an estimate.

B. Lifting, Projection and Rank Selection

For $g \in \mathbb{C}^N$, let $q = \lceil N/2 \rceil$, $m = N - q + 1$, and define

$$[\mathcal{H}(g)]_{ij} = g_{i+j}, \quad 0 \le i < m, \quad 0 \le j < q. \tag{13}$$

A channel with distinct spatial frequencies has Hankel rank at most $\min(L, m, q)$, with equality for nonzero gains and nondegenerate factors. The maximum rank is q. The inverse operation $\mathcal{H}^\dagger$ averages entries on each antidiagonal, so $\mathcal{H}^\dagger \mathcal{H}(g) = g$.

A hard-rank sweep lifts each user column, retains the largest r singular components, and averages antidiagonals. HS-GS applies Q_s such sweeps after each EM update. A finite number of these alternating steps does not guarantee a final relifted matrix of rank exactly r. We call them Cadzow sweeps rather than an exact projection onto the intersection of constraints.

The baseline uses the first 70% of pilot columns for fitting and the remaining 30% for selection. For each shared user rank $r \in \{1, \dots, q\}$, it fits the context pilots for T_{sel} updates and scores the held-out squared amplitude residual. The selected rank minimizes that score. The final estimate restarts from all pilots and runs T updates. Unless stated otherwise, $T/T_{\mathrm{sel}} = 100/25$ and $Q_s = 4$. Rank q explicitly bypasses structural processing, recovering the paired EM-GS estimate with the same initialization and updates. The option to bypass does not guarantee that the selector chooses it whenever filtering is harmful.

To distinguish selection rules, we use *HS-amplitude* for this original baseline and *HS-Rician* when the candidate score is the held-out loss in (5). Both search settings separately for each test realization; all final fits use every pilot. The shared scalar rank is a computational choice, not a claim that users have equal path counts.

C. Smooth and Data-Driven Classical Controls

Hard truncation is only one use of structure. A smooth control replaces each singular value by

$$\widehat{\sigma}_i = \frac{\sigma_i^3}{\sigma_i^2 + \tau^2}. \tag{14}$$

Its classical version selects a multiplier from $\{0, 0.05, 0.1, 0.2, 0.4, 0.8, 1.6\}$ using held-out amplitude error, with τ equal to that multiplier times the current RMS singular value. The scale is recomputed each sweep. Zero bypasses filtering. Candidate and final update budgets match HS-amplitude. This is an empirical filter, not OptShrink.

Our stronger control instead uses OptShrink weights [11] inside the same Rician EM iterations. For candidate rank $r < q$, let $s_{r+1}, \dots, s_q$ be the discarded singular values and, for $d \in \{m, q\}$, define

$$\phi_d(z) = \frac{1}{d-r}\left[\sum_{j=r+1}^{q} \frac{z}{z^2 - s_j^2} + \frac{d-q}{z}\right], \tag{15}$$

$$\mathcal{D}(z) = \phi_m(z)\phi_q(z),$$

$$\widehat{s}_i = -2\mathcal{D}(s_i)/\mathcal{D}'(s_i), \quad i \le r. \tag{16}$$

Other weights are zero. The implementation rescales the spectrum and sets weights at unresolved poles to zero. After antidiagonal averaging, a mixing factor blends the result with the EM update. Hankel lifting and nonlinear reconstruction create dependent errors; the original matrix-denoising optimality guarantees are not asserted for this adaptation.

OptShrink settings are chosen offline on independent development observations, separately by known noise level. The search includes ranks 1–15, mixing factors $0.25, 0.5, 1$, iteration budgets $10, 20, 50, 100$, and a no-filter option. Each candidate fits 21 pilots and is scored on nine withheld pilots using Rician likelihood. Settings are frozen before test generation. A final-only variant is tuned separately using the same grid. Inference uses all pilots and a noise-level lookup, without per-channel rank search. This distinction matters when comparing computation with HS-GS.

That first search covers a narrow channel population. Our broader comparison uses the same 2,048 validation observations and random 21/9 masks as the learned estimator. It adds mixing factor 0.75 and a five-update budget. The 610 combinations cover both placements, five budgets and 61 rank/mixing choices, including no filtering. Selection averages held-out likelihood equally across six path-count regimes. It freezes separate settings for each of 21 known noise/reference pairs. Path-count labels only balance offline validation; they never select settings at inference. Off-grid stress cases use the nearest lookup entry without retuning.

A final control selects rank separately for each received block. For each placement, it takes the mixing factor and update count from the broad offline lookup, fits ranks 1–15 and a no-filter option to the first 21 pilots, and selects the lowest Rician loss on the other nine. It then restarts on all 30 pilots. We test in-loop selection, final-only selection, and a joint selector over both placements. Rank is shared across users. All search and refitting work is counted, with no extra pilots. This tests observation-based rank selection; it is not an unrestricted per-block search over mixing factors, iteration counts or separate user ranks.

IV. Self-Supervised Structural Control

The learned estimator performs ten stages. Each stage fits the observations and then applies one smooth Hankel correction. The same small controller is shared across users and stages. Its interface takes only (Z, S, B, v); true channels and path counts are evaluation data, not estimator inputs.

A. Two Measurement Cores

The primary core relaxes the EM update:

$$G_{\mathrm{dc}} = G + \gamma_t(G_{\mathrm{EM}} - G), \quad \gamma_t = 2\,\mathrm{sigmoid}(a_t). \tag{17}$$

The secondary core uses $G_{\mathrm{dc}} = G - 2\eta_t D$, with

$$\eta_t = \gamma_t \frac{v}{2\|S\|_2^2}. \tag{18}$$

Both initialize $\gamma_t = 1$ and learn ten scalar update parameters. This comparison changes update geometry while retaining the exact likelihood. Validation likelihood selects the primary core before examining test channel errors.

B. Observable Controller and Structural Step

For each $H_k = \mathcal{H}(g_{\mathrm{dc},k})$, let $s_H = \|H_k\|_F/\sqrt{q}$ and $p_i = \sigma_i / \sum_j \sigma_j$. An 8–16–2 network with SiLU activation produces outputs (u, w). Its inputs are

$$\begin{aligned} f_1 &= \exp(-\textstyle\sum_i p_i \log p_i)/q, \quad f_2 = p_1, \\ f_3 &= \textstyle\sum_{i=q-\lceil q/2\rceil+1}^{q} p_i, \\ f_4 &= \log\!\left(1 + \frac{\mathrm{mean}[(Z - |G_{\mathrm{dc}}S + B|)^2]}{v}\right), \\ f_{5,k} &= \log\!\left(1 + \frac{v\,\mathrm{tr}[(SS^H)^{-1}]}{K\widehat{p}_k}\right), \end{aligned} \tag{19}$$

where $\widehat{p}_k = \|g_{\mathrm{dc},k}\|_2^2/N$, together with $P/(2K)$, $\log q$, and $(t+1)/10$. The residual mean is over all fitting observations; $t = 0, \ldots, 9$. Positive denominator guards handle zero scales. These features describe the current spectrum, residual, noise, pilot support, size and stage. During training, P here is the context-pilot count.

The outputs set

$$\tau = s_H[0.001 + \mathrm{softplus}(u)], \quad \beta = \mathrm{sigmoid}(w), \tag{20}$$

$$F_\tau(H) = H - \tau^2 H(H^H H + \tau^2 I)^{-1}, \tag{21}$$

$$g_k^{t+1} = g_{\mathrm{dc},k} + \beta_{t,k}[\mathcal{H}^\dagger F_{\tau_{t,k}}(H_k) - g_{\mathrm{dc},k}]. \tag{22}$$

Equation (21) implements (14); the matrix is normalized by s_H before the solve for numerical stability. Reconstruction gradients pass through this solve, without an identity approximation to the backward pass. Spectral features still require singular values. Threshold and gate start at approximately $0.1 s_H$ and 0.1. The network has 178 parameters; adding ten update parameters gives 188.

The implementation floors channel power and the singular-value sum at 10^{-12}, and squared Hankel RMS scale at 10^{-24}. Pilot-Gram systems use direct solves, not a hidden ridge or pseudoinverse. The older NumPy classical code uses scaled Bessel functions and a large-argument expansion above 10^4. The PyTorch learned code and both broad classical comparisons use scaled functions with a small-argument series when $\kappa^2 < 10^{-4}$. OptShrink sets a weight to zero at a relative squared spectral gap of at most 10^{-13}. These fixed numerical rules are independently checked; they do not change the observation model.

The gate and threshold can compensate for each other. Neither is an estimated rank or a guarantee of safe filtering. To measure actual intervention, we also record

$$c_{t,k} = \frac{\|g_k^{t+1} - g_{\mathrm{dc},k}\|_2}{\|g_{\mathrm{dc},k}\|_2 + \epsilon}. \tag{23}$$

In the trained models, the gate is often near one: adaptation mainly changes the shrinkage strength rather than learning a reliable on/off rule.

C. Training and Deployment

Each training realization randomly assigns 21 of 30 pilot columns to a context set A and nine to a held-out set V. Only context observations enter initialization, features and the ten updates. Training minimizes

$$\mathcal{L}_{\mathrm{hold}} = \frac{1}{N|V|} \sum_{n,p \in V} \ell\big(Z_{np}, [\widehat{G}_A S_V + B_V]_{np}, v\big). \tag{24}$$

Masks are resampled each epoch. Independent validation cases use fixed masks; checkpoints are selected by equally weighted condition means of this held-out likelihood. True channels are not used for the loss or checkpoint selection. At deployment, weights stay fixed and all available pilots enter the estimator. There is no per-channel training or rank search. Held-out likelihood remains a prediction criterion, not a guarantee of minimum channel NMSE.

Training on 21 input pilots and deploying on 30 changes the controller's input distribution, including its pilot-count feature and measurement uncertainty. This choice uses every available pilot at inference; it does not establish invariance to the split. Pilot-count transfer tests measure one consequence, but do not isolate this mismatch from the value of additional observations. A matched retraining study would be needed for that causal claim.

D. What the Updates Guarantee

For fixed τ, the singular-value multiplier in (14) lies between zero and one, so $\|F_\tau(H)\|_F \le \|H\|_F$. Also,

$$\|F_\tau(H) - H\|_F \le \sqrt{q}\,\tau/2, \tag{25}$$

since $\sigma\tau^2/(\sigma^2 + \tau^2) \le \tau/2$. Let D_H be diagonal with antidiagonal multiplicities, so $\|g\|_{D_H} = \|\mathcal{H}(g)\|_F$. Averaging is an orthogonal projection in the lifted space. Therefore the gated step does not increase this weighted channel norm, and its weighted correction is bounded by $\beta\sqrt{q}\tau/2$. These statements bound an individual output; they do not prove contraction between different inputs or reduced error relative to the true channel.

The EM surrogate is $\|Q - B - GS\|_F^2/v$. Its normal equations imply that relaxation reduces it by

$$\gamma_t(2 - \gamma_t)\|(G - G_{\mathrm{EM}})S\|_F^2/v \ge 0 \tag{26}$$

for $0 \le \gamma_t \le 2$. Subsequent structural filtering need not decrease either the surrogate or the likelihood. No convergence or low-noise consistency guarantee follows for the full learned estimator; in particular, its threshold need not vanish as measurement noise vanishes.

An NMSE-based stopping rule would require the unknown true channel and is therefore not available at inference. Rejecting a correction when likelihood worsens could protect that objective, but would not prove lower channel error or recovery of the correct phase. No such rule is added or described as a safe bypass here. The norm bound also does not make cubic shrinkage statistically optimal for the dependent errors produced by Hankel lifting. The solve is smooth for positive threshold; ordered spectral features need not be differentiable at every singular-value tie.

V. Experimental Design

The evidence comes from separate studies. Methods within a study share observations, but results from different studies are not pooled. The classical study tests structural assumptions; the original learned study tests training choices; the stronger-baseline study compares frozen learned models with tuned classical filters. Unless a fixed path count is specified, L_k is independently uniform on $\{3, 4, 5, 6, 7\}$. All estimators within a trial share the channel, pilots, reference and noise. Table I identifies the main protocols. The classical geometric generator rejects circular frequency separation below 10^{-10}, steering-matrix singular-value ratio below 10^{-8}, and pilot singular-value ratio below 10^{-8}. Reported results apply to these screened samples.

A. Learning Settings and Controls

Training uses 16,384 cases and validation uses 2,048. SNR takes seven equally spaced values from -10 to 20 dB; training RSR is 5, 12 or 20 dB. Half the cases independently sample three to seven paths per user. The other half use a common count drawn from $\{2, 4, 8, 12, 16\}$. Validation equally weights 126 SNR/RSR/path-setting combinations. Adam uses learning rate 10^{-3}, batch size 32 and 20 epochs. Training is single precision; evaluation and mathematical checks are double precision.

For each measurement core, we train no filtering (ten learned steps), globally learned filtering (one threshold and gate per stage, 30 parameters total), and the adaptive controller. Three seeds per choice give 18 models. The three seeds are 9101, 9102 and 9103. They change initialization, batch order and masks while sharing training cases. The main campaign took 3.67 hours with overlapping workers, excluding data preparation and preflight; this is not serial training time.

TABLE I
SETTINGS FOR THE PRINCIPAL COMPARISONS. $K = 3$ IN THE LISTED STUDIES; SEPARATE TRANSFER TESTS VARY K. HS BUDGETS ARE FINAL/CANDIDATE UPDATES.

Study	Array/pilots; RSR	SNR or complexity	Paired trials	Structural settings
Classical main/array	$N = 8, 16, 32$; $P = 10, 30$; 12 dB	$-5, 0, 5, 10, 15, 20$ dB	400/cell	HS: 100/25, four sweeps
Effective rank	$N = 16, 32, 64$; $P = 20$; 10 dB	Random SNR $[-10, 20]$ dB	300–400/cell	HS: 100/25, four sweeps
Classical filter comparison	$N = 32$; $P = 30$; 12 dB	$L = 3, 7, 14$; SNR 0,10 dB	400/cell	Hard/smooth: 100/25, four sweeps
Sweep sensitivity	$N = 32$; $P = 30$; 12 dB	$L = 3, 7, 14$; SNR 0,10 dB	100/cell, fresh	1,2,4,8 sweeps; own selection
Learned main study	$N = 32$; $P = 30$; 12 dB	Seven SNRs from -10 to 20 dB	400/cell	Ten stages; HS-amplitude control
Stronger shrinkage study	$N = 32$; $P = 30$; 12 dB	Seven SNRs and two stress cells	128/cell, fresh	Frozen models; tuned OptShrink
Post-review hard baseline	Same as stronger study	Same seven SNRs and stress cells	Same 1,152 worlds	One sweep; own Rician rank selection
Broad offline tuning	$N = 32$; $P = 30$; RSR 5,12,20 dB	Seven SNRs; six path regimes	4,096 fresh worlds	Same validation set; 610 candidates
Online rank selection	Same primary grid as broad study	Same grid; new channels, pilots and noise	4,096 further worlds	21/9 rank selection; all-pilot refit

The full learned evaluation has 32 conditions with 400 paired cases each. Two noise-input conditions reuse observations, giving 12,000 distinct worlds rather than 12,800 independent worlds. The stronger OptShrink comparison uses a separate development set of 224 worlds and 1,152 fresh test worlds. Neither study's error totals nor timing samples are pooled with the other. The learned models remain frozen in the stronger comparison.

After review, we add one-sweep HS-Rician to those same 1,152 saved worlds. It retains the 21/9 split, ranks 1–16 and 100/25 update budgets, but selects its own rank using one sweep in each candidate and final update. Only held-out measurements select rank. The protocol was recorded before this additional run, but the bank had already been inspected. This is a matched post-review follow-up, not a new blinded confirmation. The learned weights and OptShrink settings are unchanged.

Each broad comparison has 126 primary cells: seven SNRs, three RSRs and six path regimes (random three to seven paths per user, or fixed counts 2, 4, 8, 12 and 16). Each of two independent banks has 16 worlds per cell. Another 16 worlds per bank test each of two stress conditions: weak reference at SNR 5/RSR 0 dB and nearly noiseless observations at SNR 60/RSR 12 dB. These 64 stress worlds are excluded from the primary average. Every world redraws channels, pilots and noise. All three original learned checkpoints stay frozen; there is no new training, seed selection or ensemble. Protocols and source hashes are frozen before generating their respective test banks. Fresh test banks do not create independent training-dataset replications.

B. Error, Uncertainty and Timing

For method m and trial i, define $e_{m,i} = \|\widehat{G}_{m,i} - G_i\|_F^2$ and $d_i = \|G_i\|_F^2$. We use

$$E_m = 10 \log_{10} \frac{\sum_i e_{m,i}}{\sum_i d_i}, \tag{27}$$

$$\Delta_{\mathrm{med}}(a, b) = \mathrm{median}_i \left[10 \log_{10}(e_{a,i}/e_{b,i}) \right]. \tag{28}$$

The curve gain $E_a - E_b$ is a ratio-of-sums gain and need not equal the paired median. Positive gain favors method b. Learned summaries average per-seed dB scores and then condition scores equally. A severe error means $e_{m,i}/d_i > 1$; it is not a communication outage probability.

Intervals use 4,000 paired whole-world bootstrap resamples, retaining all users and method outputs together. For each broad study, resampling is within bank and cell, and the primary score averages the 126 cell dB NMSEs equally. Three declared macro-average contrasts use Bonferroni-adjusted 95% intervals within that study. Other intervals are pointwise unless stated. This adjustment does not cover all historical comparisons. Intervals condition on saved models and tuning settings; three training seeds do not create additional independent test worlds. We also report pooled ratio-of-sums error and rates above per-world NMSE thresholds 0.5, 1 and 2. These metrics answer different questions.

CPU timing uses one Apple M2 thread, batch size one and double precision, including initialization, feature extraction, structural steps and rank search where used. Earlier studies mix PyTorch learned code with NumPy/SciPy classical code. Both broad studies instead use PyTorch for every method; the classical implementation caches its pilot least-squares operator, while

the frozen learned code keeps its original solves. They time one independent world per condition, with warmup and three randomized-order repeats. Reported primary means exclude the two stress cells. Model loading, file access, offline tuning and training are separate. These are implementation-specific times, not device-independent speed ratios.

VI. Results

A. When Hard Hankel Structure Helps

At $N = 32$, $P = 30$ and RSR 12 dB, the paired-median HS-amplitude gain over EM-GS ranges from 2.54 to 2.97 dB across six SNRs, with mean 2.70 dB. The vertical NMSE gaps are 2.38–2.90 dB (Fig. 1). Projection is active in all 2,400 trials. At SNR 5 dB and above, EM-GS is within 0.108 dB of the Rician CRLB. The geometric CCRB is 6.95–7.06 dB below that reference, while HS-amplitude remains 4.37–4.85 dB above it. The curves do not establish efficiency or attainable headroom for these biased estimators.

The benefit grows with array size. Averaging two pilot counts and six SNRs, the paired-median gains are 0.236, 1.044 and 3.264 dB at $N = 8, 16, 32$. Corresponding ratio-of-sums gains are -0.095, 0.866 and 2.938 dB. The EM-GS average changes by only 0.120 dB. A larger array provides more spatial samples of the same path structure, whereas the unrestricted baseline also acquires more unknown coefficients. At $N = 8$, the maximum Hankel rank is only four, and projection is bypassed in 42.94% of trials. A small positive median alongside a negative total-error gain shows why both aggregation rules must be identified.

B. Effective Rank and Structural Bias

Path count alone does not describe how many components are important. For singular values of the true noiseless user Hankel matrix, define [21]

$$r_{\text{eff}} = \exp(-\textstyle\sum_i p_i \log p_i), \qquad \rho = r_{\text{eff}}/q. \tag{29}$$

For example, a spectrum dominated by two values can have lower effective rank than a spectrum with four similar values, even if both have algebraic rank four. We report the median ρ over users and worlds in a cell. This true-channel diagnostic is not supplied to either estimator.

Matched tests use $P = 20$, RSR 10 dB and SNR uniform on a millidecibel grid over $[-10, 20]$ dB. Path counts are $\{2, 4, 7, 9, 14\}$ for $N = 16$, even counts 2–16 for $N = 32$, and $\{8, 14, 29, 38, 48\}$ for $N = 64$. As ρ increases, the gain of the tested hard-rank estimator generally falls (Fig. 2). At $N = 32$, $\rho = 0.536$ gives median gain 0.141 dB with interval $[0.090, 0.211]$ dB. Thus the region near one-half of capacity marks a small benefit in this experiment, not a universal threshold.

At $N = 16$, $L = 14$ and $\rho = 0.732$, forcing rank seven gives median gain -0.154 dB, with interval $[-0.201, -0.111]$ dB. Adaptive selection has pooled median zero for these channels, but does not protect each trial. In a separate $N = 32$, $L = 14$, SNR 0 dB comparison, HS-amplitude is worse than EM-GS in 47.75% of 400 trials. Conversely, selecting below the algebraic rank need not be harmful: at $L = 7$ and SNR 0 dB, 78.75% select below seven, but only 8.75% of all trials are worse than EM-GS. Suppressing weak paths can help in noise.

C. Filter Strength and Placement Matter

The six-cell classical filter comparison uses $L = 3, 7, 14$ and SNR 0,10 dB, with 400 paired trials per cell. The smooth classical filter improves ratio-of-sums error over four-sweep HS-amplitude by 0.914, 1.382 and 1.413 dB at SNR 0 dB, and by 0.005, 0.623 and 0.773 dB at SNR 10 dB. The interval for the 0.005 dB result includes zero. This already shows that a gain over hard truncation need not require learning.

A fresh sweep study uses 100 worlds in each of the same six cells, with separate rank or threshold selection for one, two, four and eight sweeps. One-sweep hard processing improves over four sweeps in every tested cell, by 0.275–0.652 dB, with pointwise intervals above zero. More sweeps improve neither estimation nor computation automatically. These tests are separate from the learned comparison; the learned gain over the original four-sweep baseline is not evidence that it beats every hard-rank configuration.

Interleaving is also different from filtering once at the end. With the same four sweeps per application and a shared selected rank, interleaving gains 0.204 dB, with interval $[0.169, 0.242]$ dB, averaged over six per-SNR medians on 300 paired trials per SNR. With one sweep per application, the corresponding gain is 0.239 dB, with interval $[0.204, 0.282]$ dB. These comparisons hold the rank fixed using interleaved four-sweep candidates; they do not match total computation or separately tune every placement.

D. Learning Versus Simple Learned Controls

The adaptive EM estimator improves over learned step sizes alone and globally learned filtering (Table II). Its mean gain over HS-amplitude is 1.675 dB, with interval $[1.618, 1.730]$ dB; seed-specific gains range from 1.533 to 1.787 dB. At SNR 20 dB the gain is only 0.139 dB, with interval $[-0.017, 0.294]$ dB. The gradient core has slightly better test NMSE, but EM remains the primary model because validation likelihood selected it before

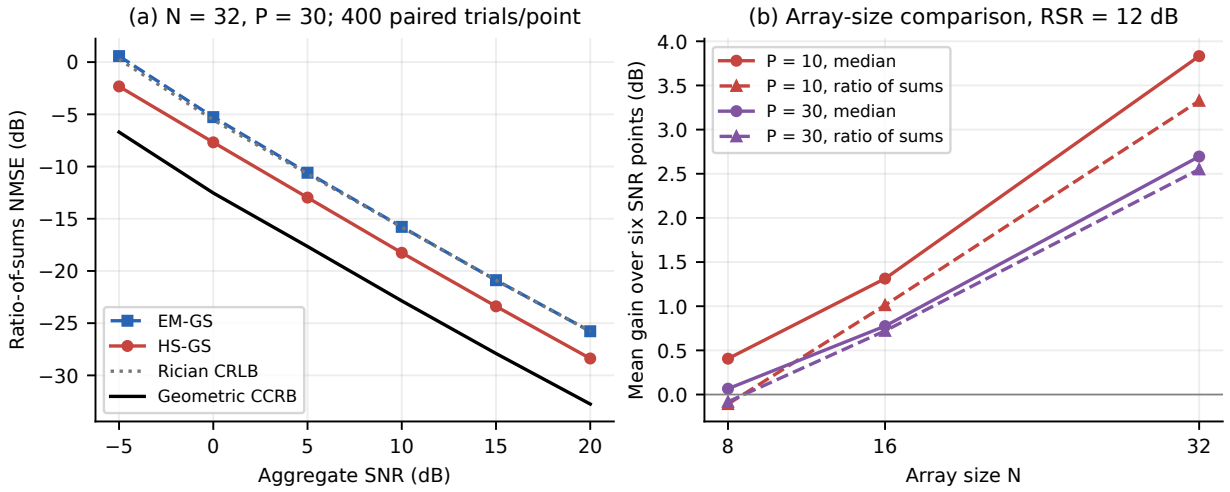


Fig. 1. Classical study. (a) NMSE and information references at $N = 32$, $P = 30$ and RSR 12 dB. (b) HS-amplitude gain versus array size for two pilot counts. The plot label HS-GS denotes HS-amplitude with four Cadzow sweeps. Panel (b) averages six SNR summaries; solid lines show paired medians and dashed lines show ratio-of-sums gains.

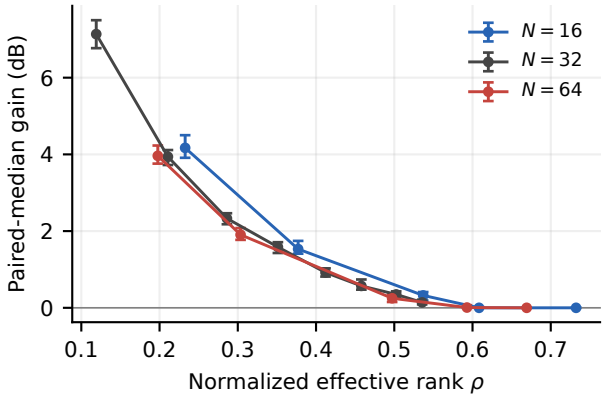


Fig. 2. Paired-median gain of the four-sweep HS-amplitude baseline against true-channel normalized effective rank. All three arrays use $P = 20$, RSR 10 dB and the same random-SNR distribution. Bars are pointwise paired-bootstrap intervals. This is a diagnostic trend, not an observable switching threshold.

TABLE II
LEARNED MAIN STUDY: SEVEN-SNR AVERAGES AND SEPARATE SNR 5 DB MEDIAN TIMING. LOWER NMSE IS BETTER.

Method	NMSE (dB)	Time (ms)	Severe (%)
Adaptive EM (188 weights)	−14.42	8.24	0.45
EM, learned steps only	−10.41	2.98	22.11
EM, global filtering	−12.43	5.61	19.19
Adaptive gradient core	−14.59	8.00	0.85
HS-amplitude, four sweeps	−12.74	525.62	14.25
EM-GS, 100 updates	−10.28	17.01	23.57
Rician gradient, hard Hankel	−11.06	47.39	20.93

test evaluation. The hard-Hankel gradient control uses development-selected rank ten and one sweep per update.

TABLE III
FRESH STRONGER-BASELINE STUDY: SEVEN-SNR AVERAGE.

Method	NMSE (dB)	Severe (%)
HS-Rician, four sweeps	−12.890	13.06
HS-Rician, one sweep	−13.314	12.95
Final-only OptShrink	−14.374	3.79
In-loop OptShrink	−14.462	5.02
Adaptive EM, frozen	−14.319	0.89

The original timing uses 100 cases and three repeats at SNR 5 dB. Adaptive EM takes median 8.24 ms versus 525.62 ms for HS-amplitude. The roughly 64-fold ratio includes full per-channel rank selection and applies only to that comparator. It does not demonstrate that learning is faster than an offline-tuned classical filter.

E. Earlier Narrow-Population OptShrink Comparison

The stronger classical control reverses the average-error ranking. In-loop OptShrink has 0.143 dB lower average NMSE than adaptive EM, with paired interval $[0.086, 0.198]$ dB (Table III). The learned method gains 0.723 dB at SNR −10 dB but loses 0.796 dB at 20 dB. Selected OptShrink uses ten iterations through SNR 10 dB and twenty at 15 and 20 dB.

The added one-sweep HS-Rician control reaches −13.314 dB, improving on four sweeps by 0.424 dB. Learning retains a 1.004 dB average gain, with paired interval $[0.922, 1.088]$ dB. This closes the one-versus-four-sweep gap on the same bank, not the comparison to every possible hard-rank method. In the separate weak-reference and two-path cells, one sweep instead beats learning by 1.976 and 1.144 dB.

TABLE IV
POST-HOC ERROR-THRESHOLD SENSITIVITY ON THE STRONGER BANK. RATES AVERAGE SEVEN SNR CELLS; POSITIVE DIFFERENCES FAVOR LEARNING.

NMSE threshold	Learned (%)	In-loop OptShrink (%)	OptShrink minus learned 95% interval (points)
0.5	17.00	15.85	$[-1.935, -0.409]$
1	0.89	5.02	$[3.051, 5.283]$
2	0.00	0.00	—

The learned method has fewer severe errors: 0.89% versus 5.02% averaged over the seven SNRs. All occur at -10 dB in this sample, where rates are 6.25% and 35.16%. Mean per-case linear NMSE favors learning, 0.197 versus 0.215, but this is a different aggregation and does not reverse the primary average-dB result. Reporting both explains the average-error versus failure-frequency trade-off without changing the primary metric after observing results.

At the original threshold $e_i/d_i > 1$, the seven-SNR reduction in failure rate versus in-loop OptShrink is 4.13 percentage points, with paired interval $[3.051, 5.283]$. At -10 dB alone the interval is $[21.354, 36.979]$ percentage points. Learned seed-specific rates there are 5.47%, 5.47% and 7.81%, versus 35.16% for OptShrink.

The conclusion depends on the error threshold (Table IV). At threshold 0.5, OptShrink has fewer exceedances; at threshold 2, neither method has any observed exceedance. Zero observed events do not establish zero population risk. These exploratory, pointwise intervals retain paired worlds and fixed models and are not adjusted for multiple thresholds. Learning therefore does not dominate the whole error distribution. The original severe-error definition and primary NMSE metric are unchanged.

The post-review timing run measures all five methods together on the same independent 24-world SNR 5 dB timing bank, with three repetitions. Mean times are 5.16 ms for one adaptive model, averaged across three trained models, 5.68 ms for in-loop OptShrink, 3.87 ms for final-only OptShrink, and 119.08/358.07 ms for one-/four-sweep HS-Rician. Initialization and setting selection or lookup are included; model loading and file I/O are excluded. These are new paired timing measurements, not averages with the earlier timing run. Backend differences remain. Learning avoids the hard method's per-case search, but has no large speed advantage over offline-tuned OptShrink.

F. Broad Comparison With the Same Validation Population

The broader study (Study A) asks a different question from the earlier RSR 12 dB, three-to-seven-path comparison: how well do the frozen methods work across the full noise/reference/path grid used for validation? OptShrink receives the same validation observations and selection loss as the learned estimator, with a larger classical search. No test-channel labels select its settings.

The compact estimator reaches -13.565 dB, compared with -13.028 dB for the validation-selected OptShrink placement. The gain is 0.537 dB, with adjusted 95% interval $[0.450, 0.695]$ dB. It also gains 0.565 dB over separately tuned in-loop OptShrink and 0.738 dB over final-only OptShrink. Table V reports all principal controls. Every original learned seed has a lower primary average than the selected classical control; seed-specific scores range from -13.327 to -13.692 dB. These are three fitted models from one training dataset, not three independent training populations.

The benefit is uneven. Learning wins in 79 of 126 cells. Averaging over noise and reference levels, it gains 2.190 dB for two paths but loses 0.272 dB for sixteen paths. Its worst primary-cell loss is 3.117 dB, at SNR 20/RSR 20 dB with sixteen paths. These results do not erase the earlier few-path losses against different controls. They show why the comparator, tuning population and averaging rule must be stated together.

Mean processing time is 4.90 ms for the compact estimator, 15.61 ms for selected OptShrink and 6.16 ms for final-only OptShrink. A separately tuned ten-update OptShrink control takes 4.56 ms and reaches -12.336 dB. Thus the learned model is not the fastest tested choice, but improves accuracy over that similar-cost control. These times count inference only; the neural model needs offline training and OptShrink needs offline tuning. The complete broad comparison, including tuning, testing, timing and audit, took about 71 minutes. That is not a separate measurement of tuning cost.

The headline gain also depends on error aggregation. Pooling linear errors across all primary worlds before taking dB reduces the gain to 0.116 dB. The learned estimator has fewer errors above NMSE one (3.03% versus 4.34%), but more above 0.5 (19.54% versus 19.12%). No errors above two occur for these two methods in this sample; this does not establish zero risk. In the excluded weak-reference check, learning loses 4.063 dB. At SNR 60 dB it reaches -6.693 dB, whereas selected OptShrink reaches -66.886 dB. The broader average supports a useful operating-range comparison, not reliable behavior outside it.

G. What Changes When OptShrink Selects Rank Online?

Study B uses 4,096 further fresh worlds. The compact estimator reaches -13.647 dB on the primary average. Online in-loop OptShrink reaches -14.154 dB, improving on learning by 0.507 dB, with adjusted 95%

TABLE V
BROAD COMPARISONS ON SEPARATE FRESH BANKS. STUDY A TESTS OFFLINE TUNING; STUDY B ADDS ONLINE RANK SELECTION ON A NEW BANK. EACH STUDY HAS 4,032 PRIMARY WORLDS AND 64 EXCLUDED STRESS WORLDS. LOWER NMSE IS BETTER. TIMES USE THE SAME PYTORCH CPU BACKEND AND INCLUDE SELECTION/REFITTING. LEARNED METRICS AVERAGE THREE FROZEN MODELS, NOT AN ENSEMBLE.

Method	A: NMSE (dB)	Time (ms)	B: NMSE (dB)	Time (ms)
Compact estimator (188 weights)	-13.565	4.90	-13.647	4.96
Offline OptShrink, selected placement	-13.028	15.61	-13.196	15.76
Offline OptShrink, in-loop	-13.000	15.57	-13.151	15.68
Offline OptShrink, final-only	-12.827	6.16	-12.863	6.23
Online rank + placement selection	—	—	-14.096	114.17
Online rank, in-loop	—	—	-14.154	108.70
Online rank, final-only	—	—	-13.723	13.96
HS-Rician, one sweep	-12.996	98.91	-13.059	99.88
EM-GS, 100 updates	-9.922	13.08	-9.958	13.15

interval $[0.437, 0.562]$ dB. Joint rank-and-placement selection reaches -14.096 dB and improves on learning by 0.449 dB, with interval $[0.358, 0.569]$ dB. Final-only rank selection reaches -13.723 dB; its 0.076 dB point advantage has interval $[-0.025, 0.226]$ dB. That interval establishes neither superiority nor equivalence.

The in-loop selector is therefore the strongest tested method on this primary accuracy measure. The joint selector was declared in advance and is reported even though searching more choices did not give the lowest test error. Rician prediction loss on nine withheld pilots is not a channel-error guarantee, and restarting on all pilots changes the fitted estimate. We do not identify either effect as the sole cause of the observed ranking.

The cost is substantial in this implementation: mean times are 108.70 ms for online in-loop selection, 114.17 ms for joint selection, 13.96 ms for final-only selection and 4.96 ms for the compact model. All candidate fitting and all-pilot refitting are included. Inference is about 22 times faster for the learned model than for online in-loop selection under this single-thread, batch-one test. This ratio does not include amortized training cost and is not a prediction for other devices, batch sizes or optimized implementations.

Learning has fewer primary errors above NMSE one: 3.08%, compared with 6.87% for online in-loop and 5.78% for joint selection. At threshold 0.5, learning has 19.55% exceedances, compared with 19.52% and 19.22%, respectively. Thus the primary accuracy winner and the winner at one severe-error threshold differ. Neither method's settings were selected to minimize that error frequency, and these observations are not communication-outage probabilities.

Pooling linear errors across primary worlds also favors learning: its pooled NMSE is -6.683 dB versus -6.217 dB for online in-loop selection. This descriptive difference does not replace the declared average-cell-dB result. The former is more influenced by large errors; the latter gives each operating cell equal weight in dB.

The limits remain large. The compact model's worst primary-cell loss to the in-loop selector is 5.495 dB. Joint selection also beats it in the weak-reference stress case (-10.101 versus -7.787 dB) and the nearly noiseless case (-70.134 versus -6.835 dB). These failures remain outside the primary average but inside the report. The final result supports a measured accuracy–cost trade-off; it does not support the claim that our estimator beats tuned OptShrink once observation-based rank selection is allowed.

The failed extensions are also relevant. A corrected learned gate around OptShrink gains only 0.019 dB over its strong starting baseline and loses to retuned classical mixing and a simpler stage-only learned rule. Simple stopping/filter-release rules do not improve their primary average. A small-budget exact-EM Transformer run is worse than EM100 and is not a strong URformer comparison. These separate pilots do not support replacing the original compact estimator or claiming a new improved OptShrink algorithm. Their results and the failed local error-model screen are retained in the supplement.

H. Operating Limits and Additional Checks

Against HS-amplitude, learning loses 1.617 dB at RSR 0 dB, 1.104 dB for two paths and 0.203 dB for three paths. These losses prevent a claim that the controller reliably identifies every channel on which filtering helps. The stronger study also finds no universal winner. At RSR 0 dB, selected OptShrink reaches -5.026 dB versus -7.069 dB for learning, but longer 50- and 100-update OptShrink runs reach -8.790 and -8.981 dB. Transferring a noise-dependent stopping rule can fail when reference strength changes. At two paths, HS-Rician reaches -17.833 dB, learning -16.814 dB and selected OptShrink -16.407 dB.

Input-removal tests train on 4,096 cases for 12 epochs with three seeds and evaluate two fresh banks totaling 2,304 worlds. Full inputs improve average error

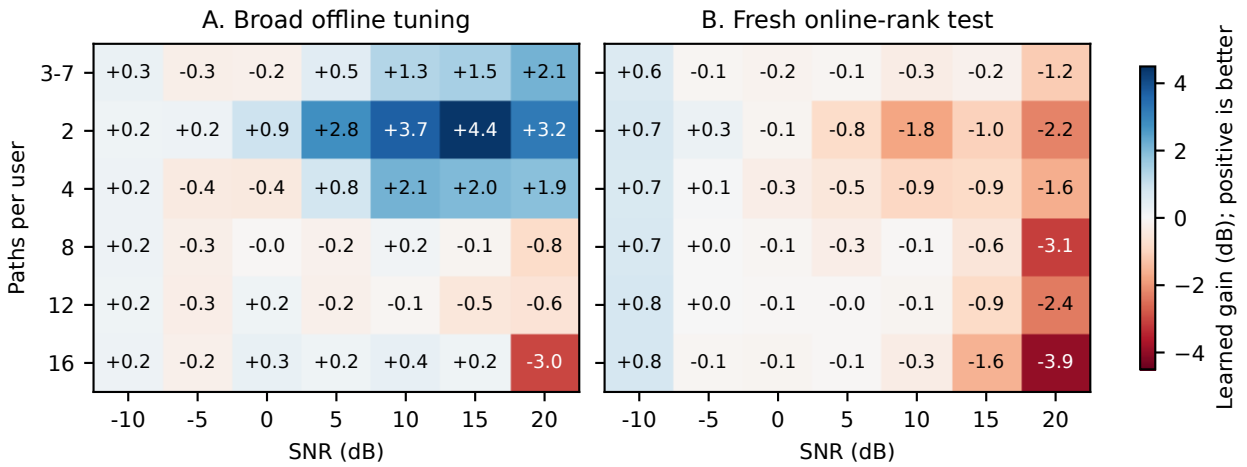


Fig. 3. Learned gain over (a) broad offline-selected OptShrink and (b) online in-loop rank selection on a separate fresh bank. Positive values favor learning. Each square averages three RSR-specific cell dB gains; the random 3–7 row draws path counts independently for each user. These descriptive averages are not cellwise confidence guarantees. The full individual-cell values are retained in the evidence records.

over spectrum-only, noise/residual-only and size/stage-only controllers. However, they lose 0.39–0.40 dB to noise/residual-only inputs at SNR 5 dB and 0.82–0.93 dB on two-path cases. Both information groups can help the average without improving every condition. A separately trained final-only learned filter is about 0.70 dB worse on its follow-up study's declared average; that experiment has its own training and test data and does not replace the classical shared-rank placement comparison.

Frozen-model dimension tests cover $N \in \{8, 16, 32, 64\}$, $K \in \{2, 3, 4\}$ and $P \in \{10, 20, 30\}$ one factor at a time. The gain interval at $N = 8$ includes zero. A smaller fresh pilot-count study, with 32 worlds per cell, extends to $P = 6 = 2K$. At SNR 5 dB, learned NMSE is -2.197 dB with six pilots and -14.332 dB with thirty. Full local Jacobian rank in sampled nonzero-reference cases does not ensure useful finite-noise accuracy.

Array mismatch also affects structural benefit. In a separate classical check at $N = 32$, $P = 30$, SNR 5 dB and RSR 12 dB, independent Gaussian element-position errors increase in standard deviation from zero to 0.02λ. The reference remains calibrated and error is measured against the perturbed channel. HS-amplitude gain drops from 4.955 to 3.917 dB for three paths, and from 0.264 to 0.050 dB for twelve paths; the latter interval includes zero. These are model-mismatch simulations, not hardware measurements.

A nearly noiseless check is especially instructive. At SNR 60 dB, outside the training range, and RSR 12 dB, the learned model reaches -6.647 dB versus -65.715 dB for EM-GS with 100 updates on 32 worlds. Disabling only filtering gives -44.020 dB; starting the full learned model at the true channel still gives -6.653 dB. These post-hoc diagnostics show that filtering itself can spoil an accurate estimate. They do not define a validated switching policy.

I. Retained Training and Calibration Evidence

A separate reduced-budget study trains 25 models for 300 epochs in total and tests 3,584 fresh worlds across 56 SNR/reference/path conditions. Its training mixture includes RSR 0 dB, unlike the original study. Relative to its ten-stage controller, five stages lose 1.977 dB, while twenty gain only 0.181 dB and nearly double mean processing time. Increasing width from 16 to 32 gains 0.065 dB; two experts gain 0.023 dB; curriculum training gains 0.206 dB but loses 2.353 dB in its worst cell. No variant improves every cell. Three independent training datasets give two-initialization baseline means of -12.450, -12.124 and -12.229 dB. These are limited replications, not precise population uncertainty. The original primary models are not replaced using these test results.

Calibration sensitivity was also measured, not only discussed. In a 32-world SNR 5 dB check, assumed constant reference-phase errors of $\pm 10^\circ$ worsen learned NMSE by 2.36–2.47 dB; ramps from zero to $\pm 10^\circ$ worsen it by 0.78–0.84 dB. Supplying variance 6 dB too high causes a 3.05 dB loss. A separate 192-world diagnostic estimates a shared positive reference gain and noise variance while keeping phase supplied. It corrects some amplitude errors but worsens one tested cell by 1.048 dB, and adds 45.05 ms of preprocessing. It does not resolve unknown phase or drifting atomic response. The accompanying supplement retains these protocols,

full tables, feature ablations and selected/final checkpoint checks; none is pooled with the primary comparison.

VII. Discussion and Conclusion

The two parts of the study lead to the same practical lesson: structural processing must be assessed against the error it removes, the bias it introduces and the work required to choose it. Low effective rank helps explain hard-projection gains, but effective rank is not an observed receiver input here. A rank-selection bypass, a learned gate and norm-nonincreasing filtering each fall short of a per-trial protection guarantee.

The learned method is most relevant when repeated estimation near a known operating population can reuse offline training. It improves on global learned filtering and broad offline-tuned OptShrink, but observation-based in-loop rank selection has lower average cellwise error. The compact model avoids that search and has lower measured processing time. It also has fewer errors above NMSE one, without dominating the full distribution. Thus a network is not necessary for the best tested primary accuracy. Neither method is globally optimal or safe on every channel. The older narrow-population win for OptShrink remains valid; different populations and protocols should not be pooled into one headline score.

For $m = N - q + 1$, one adaptive stage costs $O(NKP + NK^2 + K^3 + Kmq^2 + Kq^3)$ in the present implementation. Features still require an SVD, and reconstruction requires a dense solve. HS-GS repeats structural operations across candidate ranks and updates. A smaller parameter count alone is therefore not evidence of lower computational cost. Nor does applying the structure more often necessarily help: both the sweep study and the separately tuned final-only OptShrink control argue against that assumption.

The scope remains simulation-based estimation under an effective magnitude model with calibrated reference. The main confidence intervals exclude training-dataset uncertainty. Independent-data replications are limited; the new one-sweep comparison reuses an inspected bank. No matched URformer, LRPR-VBL, HVaF or full LRHD experiment is reported, so their accuracy and runtime relative to this controller are unresolved. There is no global recovery or full-estimator convergence theorem and no hardware validation. Non-Gaussian noise, outliers, two-dimensional arrays, wideband propagation and unknown phase drift are outside the tested model. Noise-variance mismatch alone is not a test of robustness to a different noise distribution.

We have presented classical and self-supervised Hankel estimation within a common Rician framework. Hard structure helps when the channel leaves enough unused Hankel capacity, but rank selection and repeated projection can introduce avoidable error. A small controller learns useful smooth filtering without channel targets and avoids per-channel search. It improves the broad offline-tuned comparison, but loses average cellwise accuracy to online in-loop OptShrink rank selection. Its measured advantage is lower processing cost and fewer errors above the declared severe-error threshold, subject to clear operating limits. This is not universal superiority, a hardware demonstration or a full-estimator convergence guarantee.

A. Reproducibility

The supplement records numerical details, the two broad protocols and retained negative follow-ups. Local reproduction packages contain generation seeds, observations, checkpoints, source hashes, predictions and audit scripts. The broad offline and online-rank studies each regenerate all 4,096 test worlds and independently recalculate headline statistics. Representative predictions are replayed, including independent numerical checks of OptShrink scores and refits. This is not a replay of every historical run. The manuscript archive contains source, figures and compact evidence records; full experimental data are separate. A permanent public code/data destination remains to be assigned before submission.

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